

Missing data imputation using an iterative denoising autoencoder (IDAE) for dissolved gas analysis

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ABSTRACT

With the expansion of the energy market, safe and stable operation of the electrical power system has become an important issue. In an effort to achieve this goal, much research has been conducted on diagnosis approaches suitable for core components of the electrical power system. Transformers are one such core component. Most of the research on transformers has focused on developing a diagnosis model; less effort has been devoted to the data, in spite of the fact that such models require data of sufficient quantity and quality, which is not usually readily available for transformers. Thus, in this paper, we propose a way to fully exploit the valuable transformer data, using a data imputation approach called the iterative denoising autoencoder (IDAE) method. The proposed method imputes missing values of dissolved gas analysis (DGA) data, which is frequently lost, for various reasons. IDAE can help diagnose the health state of transformers accurately by estimating the missing values of DGA data. The proposed method is verified in this research through three comparative studies that examine field data provided by an electric power corporation. Specific studies provide: (1) a comparison with conventional methods on imputation performance for a single gas, (2) examination of imputation performance between multiple missing values, and (3) documentation of diagnosis accuracy before and after imputation. The results of the case studies show that the proposed method is effective for imputation of the missing DGA data.

1. Introduction

As market opportunities related to smart grids and sustainable electric networks have grown, concerns about the stability of the electric power system have increased. Transformers, one of the main components of the electric system, need careful management. However, as transformers are used over several decades, they degrade and become subject to abrupt accidents that arise for various reasons, such as abnormal voltage, careless operation, and insulation degradation. To avoid transformer failure, global standard organizations, such as IEEE and IEC, suggest maintenance guidelines based on the domain knowledge.

Much research has been conducted towards the goal of accurately diagnosing and maintaining transformers. amongst prior methods, dissolved gas analysis (DGA) is the most widely used. When the insulator of a transformer is dismantled by thermal and electrical stresses, combustible gases are generated and dissolved into the oil that is filled inside transformers. The amount and the ratio of these dissolved gases

depend on the degradation condition of the transformer. Therefore, the condition of the transformer can be estimated through gas analysis. There are several global standards on dissolved gas analysis, such as IEEE Std C57.104 [1], IEC 60,599 [2], Duval Triangle [3], Doernenburg Ratios [4], Rogers Ratios [5], and Basic Gas Ratios [2]. These methods have been improved based on decades of study; however, they still have low accuracy and a high frequency of false alarms.

As transformer data has accumulated over long periods, and techniques based on big data analysis and artificial intelligence are maturing, much research has been conducted with the aim of inferring the status of a transformer using DGA data. In the early stages of these methods, algorithms using machine learning, including fuzzy logic [6–8], artificial neural networks [9–13], and support vector machine [14–19], are often applied. Recently, state-of-the-art technologies that incorporate deep learning have been adopted to construct diagnosis models for transformers [20–23]. For example, L. Luo et al. [22] proposed a DGA online fault diagnosis method that combines a convolutional neural network and a bidirectional long short term memory

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network. In other work, D. Yang et al. [23] developed a double-stacked autoencoder for fast and accurate judgement of transformer health condition.

Although artificial intelligence methods can greatly enhance the accuracy of diagnosis, and verify various failures of a transformer, they require a large amount of and high-quality data to construct a robust model. Unfortunately, in the case of transformers, obtaining such data is difficult. Transformer data is rarely measured, usually only once or twice a year. Further, the data may be corrupted for a variety of reasons. The quality of DGA data greatly depends both on the techniques used to extract gas from the insulating oil and the skill of the personnel gathering the data. According to research by Cho et al. [24], the error rate of the round-robin test for each laboratory is 15–30%, on average. In addition, Dukarm reported common issues arising from DGA data, such as low measurement precision, contradictory data, and intermittent gas losses, all of which require special attention when measuring DGA gases [25].

To supplement the insufficiency of data in both quantity and quality, various preprocessing methods have been developed. Data generation methods such as data augmentation and data resampling are active ways of solving the quantitative problem. One prominent method is the generative adversarial network, proposed by Goodfellow et al. [26]; many other methods have been derived from it. Several studies have also been pursued on DGA data of power transformers; these approaches use data generation methods to overcome the data imbalance between normal and fault states. V. Trad and Y. Cui [27,28] developed an adaptive over-sampling method. S. Kim, et al. [21] overcame the unlabelled data problem by using a semi-supervised autoencoder (SSAE). These methods enable diagnosis of various states of the transformer, as well as enhancement of the accuracy of the diagnosis.

To make the most use of the data, other preprocessing methods have been proposed, with the goal of increasing the quality of the data. A representative preprocessing method of this kind is noise reduction. Most sensory data contain noise signals from external sources or operating environments. For accurate diagnostics, clean signals, from which noise has been removed, should be extracted. Ha et al. [29] proposed autocorrelation-based time synchronous averaging (ATSA) to isolate a vibration signal of a planetary gearbox from noisy sensory signals. ATSA designed a new window function with optimal size and shape to identify a vibration pattern similar to that of a planetary gear. Eom et al. [30] proposed an improved Kalman filter to reduce noise and obtain refined data in multi-sensing systems. By optimizing the noise covariance parameters of the Kalman filter approach, more accurate data can be obtained from smart RFID tags.

Another type of preprocessing is data imputation, which recovers missing values from incomplete data. Zariah et al. [31] proposed the FINNIM model, which is a nonparametric iterative imputation method that is based on the k-nearest neighbors (KNN) approach. FINNIM prioritizes the gas that needs to be corrected based on the Fisher relationship between the dissolved gas and fault type. This approach imputes the missing values by performing KNN iteratively until convergence. Lin et al. [13] suggested the DGA imputation algorithm, which leverages KNN and XGBoost. The methodology is similar to FINNIM; however, it obtains the relationship between the dissolved gas and fault type through an XGBoost classifier and imputes the missing value through KNN. Both FINNIM and XGBoost use KNN as the imputer, which is limited in accuracy based on the number of the nearest neighbors and the distance function.

In this study, we propose a new imputation model that is based on the iterative denoising autoencoder (IDAE) approach; this approach is designed to overcome the limitations of KNN-based methods. The proposed method consists of three steps: (1) defining the inputs and data normalization, (2) DAE model learning for a single missing value, and (3) IDAE for multiple missing values. The main idea of this study is to enable DAE imputation model for estimation of the original data to replace the incomplete data with multiple missing values by repeating

the imputation process until the missing values converge. The proposed method is verified using DGA data obtained from more than 8000 transformers installed at actual industrial sites, provided by Korea Electric Power Corporation (KEPCO). The key features of this study can be summarized as follows.

- By using a neural-network-based imputation model, accuracy is improved, as compared to conventional imputation methods, such as KNN or decision-tree-based models.
- By imputing the various and multiple missing values, more accurate and robust transformer diagnosis is possible.
- Since a large amount of DGA data acquired in actual industrial sites is used, not only does the study have academic value, but it can also be applied to real industrial problems.

The remaining parts of this paper are organized as follows. Section 2 introduces the failure modes of a transformer and provides an overview of the traditional DGA methods used to diagnose the status of a transformer. Section 3 explains the theoretical background on the DAE approach and how DAE is used in this study. In Section 4, the proposed method is verified by studying transformer field data acquired from industry. Finally, the conclusions and suggestions for future research are provided in Section 5.

2. Transformer diagnostics based on DGA

This section first describes the possible failures of a transformer and their causes. Then, dissolved gas analysis is explained as a diagnostic method for such failures. Lastly, data issues that arise in dissolved gas analysis are introduced.

2.1. Failures of transformers

Transformers are one of the most important components in the electric power system. Transformers connect power systems with different voltage levels. As a transformer experiences enormous electrical and mechanical stresses as it operates under high-voltage conditions, internal components (e.g., insulating paper and windings) undergo degradation. This degradation ultimately results in the abrupt failure of the transformer, which interrupts the power grid that is connected to the transformer.

A transformer can fail by electrical, thermal, and/or mechanical defects. Electrical defects result from transient over-voltage or winding resonance. Thermal defects may arise from an overload current, local overheating, leakage fluxes and/or failure of the cooling system. Mechanical stress between the conducting material and the winding occurs because of a short circuit of the winding and inrush current.

Such defects in a transformer are caused by a decrease in the mechanical and dielectric strength of the transformer as the internal insulator ages. In general, aged conductor insulation is weakened to the point where it can't withstand the mechanical stresses of a fault. Indeed, the insulation becomes so brittle that even normal operating conditions may cause severe damage. Then, dielectric failure of the turn-to-turn insulation or loosening of the winding clamping pressure occur, which reduces the transformer's resistance to short-circuit forces [32]. Typical major insulation defects occurring in the transformer are as follows: moisture in the cellulose insulation, contamination of oil with water or particles, insulation surface contamination (which occurs mainly due to the adsorption of polar ageing products on a cellulose surface or due to deposition of conducting particles and insoluble ageing products), and partial discharges in weaker parts of the insulation [33].

2.2. Dissolved gas analysis

As previously described, there are various failure modes of transformers; thus, developing a diagnosis model for each failure would be

costly and ineffective. However, most failures are the result of the generation of dissolved gas in the oil filled inside a transformer. Thus, by analysing the dissolved gas, the overall status of the transformer can be monitored, and the failure can be predicted in advance and thus prevented. The examination of dissolved gas is commonly called dissolved gas analysis (DGA).

A high-voltage transformer is usually insulated with insulation paper and oil to safely handle the high voltage. Two levels of insulation are used. First insulation paper is wrapped around the coil and iron core of the transformer. Next, the inside of the transformer where the coil and iron core are located is filled with oil for further insulation. When there is local heating, arc, or thermal or electrical stress, the insulation material is decomposed and combustible gases (e.g., H_2 , C_2H_2 , C_2H_4 , C_2H_6 , CH_4 , CO , CO_2 , N_2 , O_2 , and C_3H_8) are generated. The ratio of these gases varies according to the cause of the decomposition. Therefore, DGA is usually utilized to analyse the gas ratio and to infer the status of the transformer.

Various diagnostic methods can be used with DGA. The Duval [3] method may be the most well-known method. It measures the gas ratios of CH_4 , C_2H_2 , and C_2H_4 , and determines three types of faults based on these levels. Fault types include thermal faults, electrical faults (discharge of low and high energy), and partial discharge. Another representative diagnosis method is IEEE Std C57.104 [11]. This method measures the concentrations of seven gases, H_2 , C_2H_2 , C_2H_4 , C_2H_6 , CH_4 , CO , CO_2 , and the sum of all combustible gases (TDCG) to determine the status of the transformer. In this method, the state is classified into four severity levels. In addition to these methods, there are many other methods that have been proposed based on dissolved gas analysis.

2.3. Data issues in DGA

Although DGA is widely used due to its simplicity and generality in inferring the various failures of a transformer, a problem arises when this method is applied in practice; specifically, it has difficulty gathering reliable data. Dissolved gas data is subject to corruption for various reasons. This is described in reference [25], which notes the importance of careful data analysis. Some cases of unreliable data are as follows. First, the gas ratio can be significantly different between samples. The samples obtained at similar times should have similar quantities; if not, the samples should be considered to be error. A low concentration of H_2 or CO_2 also indicates possible error. Because the solubility of these gases is very low, the oil rapidly loses these gases when exposed to the air. Therefore, a low ratio of H_2 or CO_2 means that the sample is corrupted by exposure to the air. Similarly, an increase in O_2 and N_2 occurs during the exposure, because these gases are abundant in the air and readily dissolved in the oil.

The causes of unreliable data vary. Most of these causes stem from careless handling of samples. In many cases, the samples are manually gathered by workers. While taking the samples, it is easy for the samples to be exposed to the air if they are not completely sealed. Also, the gases can be lost when the transformer has a small crack or rupture in the transformer housing. Some minor cases of unreliable data include mislabelling, data transcription error, and incorrect analysis. Unreliable data should be excluded from the analysis. However, transformer data is not frequently obtained; it is typically measured only once or twice a year. Therefore, every data sample is valuable and should be fully exploited if possible, rather than being excluded due to reliability concerns. To this end, the proposed method is developed to enhance the useability of unreliable or incomplete data.

3. Proposed method for missing DGA data imputation

This section describes the proposed method for estimation of the missing DGA data using a denoising autoencoder. The DAE method is first reviewed, then the structure of the DAE model for use with transformer data is explained.

3.1. Denoising autoencoder

To estimate the missing values in a transformer's DGA data, a denoising autoencoder (DAE), first proposed by P. Vincent and Y. Bengio, is adopted [34]. DAE is the expanded version of an autoencoder, which is used to recover the original data from noise-corrupted data. DAE is based on the fact that the data maintains its essential characteristics, even when partially destroyed. Therefore, the DAE model can recover the original data from the noise-added input. DAE is widely used for image/voice recovery, typo correction, and noise filtering, amongst other applications [35–40].

The structure of the DAE model is shown in Fig. 1. It consists of two parts. The first part is the autoencoder, which itself can be decomposed into encoder and the decoder parts. The role of the encoder is so-called manifold learning, which sequentially reduces the dimension of the input data. As a result, the essence of the original data, called the latent values, which contains enough information about the original data, is obtained. For given data $\mathbf{x} \in R^D$, the encoder function, $\mathbf{f}_\theta$ is expressed as shown in Eq. (1).

$$\mathbf{f}_\theta(\mathbf{x}) = \mathbf{h} = s(\mathbf{W}\mathbf{x} + \mathbf{b}) \quad (1)$$

where $\mathbf{W}$ is the $d \times D$ dimensional weight matrix, $\mathbf{b}$ is the d dimensional bias vector, $\mathbf{h}$ is the d dimensional latent value, and $s(\cdot)$ is the activation function.

The decoder is the reverse process of the encoder, which is called generative model learning. The decoder uses sequentially increasing layers, and recovers the original data from the output of the encoder. The equation for the decoder, $\mathbf{g}_{\theta'}$ is as follows.

$$\mathbf{g}_{\theta'}(\mathbf{h}) = \hat{\mathbf{x}} = s(\mathbf{W}'\mathbf{h} + \mathbf{b}') \quad (2)$$

where $\hat{\mathbf{x}} \in R^D$ is the recovered data from the input of the decoder or, equivalently, the encoder of output, $\mathbf{h}$, and weight parameter $\theta = \{\mathbf{W}, \mathbf{b}\}$ and $\theta' = \{\mathbf{W}', \mathbf{b}'\}$ are estimated through the learning process.

The autoencoder learns the weight parameter θ and θ' by minimizing the loss function $L(\theta, \theta')$, which measures the similarity between $\mathbf{x}$ and $\hat{\mathbf{x}}$. In this study, the mean-square error function of Eq. (3) is adopted as a loss function to estimate the missing values. Given the training dataset $\{\mathbf{x}^1, \dots, \mathbf{x}^N\}$, the loss function is minimized by updating θ and θ' through a backpropagation method that is based on the gradient descent algorithm [41].

$$L(\theta, \theta') = \frac{1}{N} \sum_{k=1}^N \|\mathbf{x}^k - \hat{\mathbf{x}}^k\|^2 = \frac{1}{N} \sum_{k=1}^N \|\mathbf{x}^k - \mathbf{g}_{\theta'}(\mathbf{f}_\theta(\mathbf{x}^k))\|^2 \quad (3)$$

The other part of DAE is the addition of noise to the raw data. The noisy data after this step is designated as $\tilde{\mathbf{x}}$ in Fig. 1. The noisy data is generated using the stochastic corruption procedure $\tilde{\mathbf{x}} \sim q_D(\tilde{\mathbf{x}}|\mathbf{x})$ [42]. Through this process, about half of the input data is randomly substituted with zero. Noisy data is then provided to the autoencoder model, which generates $\hat{\mathbf{x}}$. Since the loss function is defined as the error between the original data $\mathbf{x}$ and the reconstructed data $\hat{\mathbf{x}}$, as the training goes on, the model takes noisy input $\tilde{\mathbf{x}}$ and gives an output similar to original data $\mathbf{x}$. In other words, the model now acts like a noise filter.

3.2. DAE-based iterative imputation for multi-missing DGA

The overall process of the proposed method is shown in Fig. 2. It consists of three main steps: data preprocessing, DAE training, and IDAE. In the data preprocessing step, the input dimension is defined, and normalization is performed. In the DAE training step, a DAE model to recover a single missing value is constructed. Then, IDAE, an imputation model for the multiple missing value situation, follows. The initial values for the multiple missing values are first assigned by KNN, and the IDAE updates the initial guesses to robust values. In the following

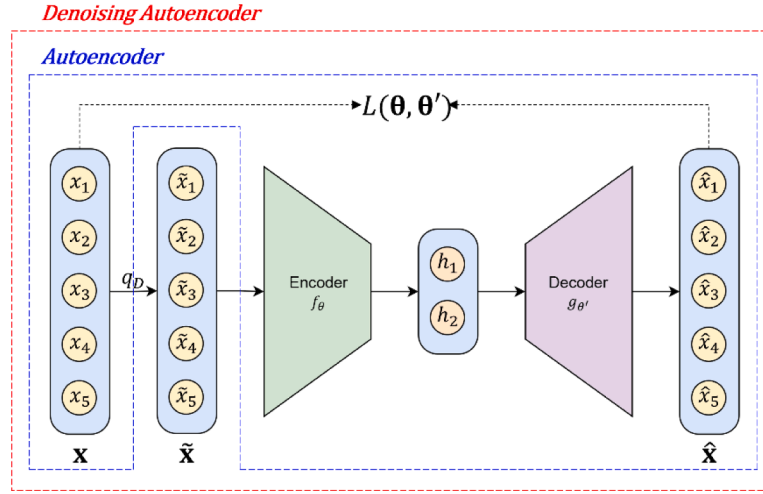


Fig. 1. The structure of the DAE model.

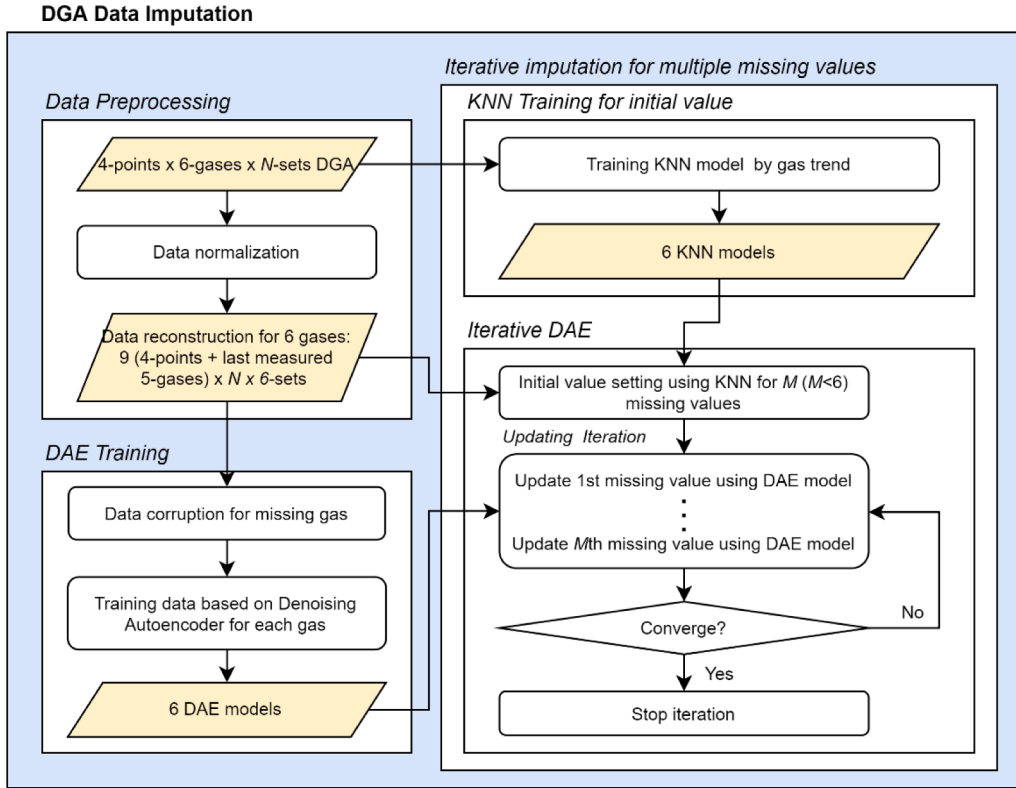


Fig. 2. Overall process for missing data imputation in DGA.

subsections, each step is described in detail.

3.2.1. Data preprocessing

First, to apply the DAE model to the transformer data, the input needs to be defined for accurate reconstruction of the missing data. One way of doing this is to use the time-trend of the DGA data. As the gases are accumulated over time, the amount of each gas monotonically increases. Hence, the missing data may have a value in between the values before and after the missing data. However, DGA data is generally not measured often (i.e., once or twice a year); thus, a gradual change sometimes is not clearly seen because of interruptions by other electrical events. For example, sudden partial discharge could rapidly increase the amount of gases. Therefore, when we estimate the missing gas

concentration only using time-series trend information, it will not be able to interpret the case where the gas has abrupt change.

Thus, in this study, in addition to the time-trend data, the gas concentrations are also used to facilitate the determination of the missing value at a certain time. The amount of a gas can be estimated using the other gases, when the failure mode is verified. As a result, the input of the DAE model is designed to include both the time-series data of a gas and the amounts of other gases at a specific time, t . The input is defined as follows.

$$\mathbf{x} = [x_{t-3}^i, x_{t-2}^i, x_{t-1}^i, x_t^i, x_{t+1}^i, x_{t+2}^i, x_{t+3}^i, x_{t+4}^i, x_{t+5}^i] \quad (4)$$

where the superscript and subscript of x are gas type and measurement time, respectively. In Eq.(4), x_t^i represents the missing value and will be

estimated using both the time-series data, $[x_{t-3}^i, x_{t-2}^i, x_{t-1}^i]$, and gas concentration at time t , $[x_t^i, x_{t+1}^i, x_{t+2}^i, x_{t+3}^i, x_{t+4}^i, x_{t+5}^i]$. The six gases include H_2 , C_2H_2 , C_2H_4 , C_2H_6 , CH_4 , and CO . The missing value x_t^i is set to be -0.1 , which is physically impossible to obtain, so the learning algorithm can recognize it.

Another preprocessing step is normalization. The scales of the dissolved gases are different. Some gases are more easily generated and dissolved in the oil than other gases because of their low enthalpy. In contrast, some gases such as C_2H_2 require high enthalpy to be formed. In this case, the gas is generated only when there is a fault that releases high energy such as arc or corona. Because this high-energy condition is not readily met, C_2H_2 shows rather low concentrations in the oil. However, a fault with high energy indicates a possible severe fault; thus, it is an important indicator regardless of low concentrations in the oil. Thus, to reduce the scale difference, min-max normalization is adopted, which transforms the data to have a value between 0 and 1.

3.2.2. DAE model for a single missing value

Using the input defined previously, the model is trained. Before dealing with multiple missing values, a model for a single missing value is first trained for each gas, based on DAE. As a result, a total of six models are obtained. The structure of the DAE model is shown in Fig. 3. In the input layer, the data marked 'x' represents the missing data. Each input consists of nine elements, including the time trends and gas concentrations, as mentioned earlier. The corrupted inputs are reconstructed through the encoder and decoder layers. The encoder and decoder layers consist of five hidden layers. Since the encoder and decoder have low input dimension and all input elements are meaningful, the layers are designed to have a fully connected layer denoted as 'fc', as shown in Fig. 3. The number of nodes of each hidden layer are set to 40, 20, 10, 20, and 40, respectively. Every layer except the last uses the exponential linear unit (ELU) function as its activation function, as shown in Eq. (5). The last layer uses the hyper-tangent function described in Eq. (6) to let the input data fall between 0 and 1. Detailed parameters of the DAE model are summarized in Table 1. The number of epochs and batch size are set as 1000 and 100, respectively. The learning rate η is set as 0.001. The output through the auto-encoder has recovered value, which is denoted as 'o' in the figure. The mean square error of Eq. (3) is adopted for the loss function, and training is conducted to minimize the loss function. This trained model is only applicable for data with one missing value. However, many DGA data include multiple missing values. Thus, an iterative method for multiple missing values is proposed in the following section.

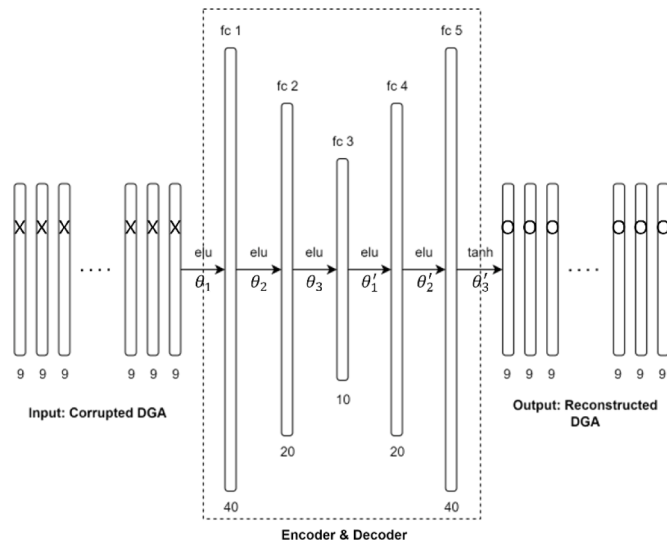


Fig. 3. Example of the DAE model structure for one gas.

Table 1

Parameters in the structure of the proposed DAE.

Layer	Activation	The number of nodes	The number of parameters
Input	–	9	–
Encoder 1	ELU	40	400
Encoder 2	ELU	20	820
Encoder 3	ELU	10	210
Decoder 1	ELU	20	220
Decoder 2	ELU	40	840
Output	tanh	9	369

$$R(x) = \begin{cases} x & x \geq 0 \\ \alpha(e^x - 1) & x < 0 \end{cases} \quad (5)$$

$$\tanh = \frac{1 - e^{-x}}{1 + e^{-x}} \quad (6)$$

3.2.3. DAE-based iterative imputation for multiple missing values

The previous DAE model works only for a single missing value. To expand the previous DAE model to data with multiple missing values, the missing values are initially estimated and iteratively updated until they converge. For initial estimation, the k-nearest neighbourhood (KNN) method is adopted, because it shows quite high accuracy when there is enough data whose dimension is small. In addition, the learning method is simple and requires little effort to estimate parameters. It is important to set the optimal initial value carefully because, by adopting KNN, it is possible to avoid a situation in which the initial estimation falls into the local minimum or plateau during estimation.

The KNN method estimates the target value by finding samples that are similar to the target data. If the target data is $x_{t,0}^i$, which is the i th gas at time t , the input to the KNN model is three time-series data before $x_{t,0}^i$, which is represented as $\mathbf{z}_0^i = [x_{t-3,0}^i, x_{t-2,0}^i, x_{t-1,0}^i]$. Then, the distances between $\mathbf{z}_0^i$ and the j th sample $\mathbf{z}_j^i = [x_{t-3,j}^i, x_{t-2,j}^i, x_{t-1,j}^i]$ are measured as follows.

$$d_j^i(\mathbf{z}_0^i, \mathbf{z}_j^i) = \sqrt{(\mathbf{z}_j^i - \mathbf{z}_0^i)(\mathbf{z}_j^i - \mathbf{z}_0^i)^T}, \quad (j \neq 0) \quad (7)$$

Finally, K samples of $\mathbf{z}_k^i$ ($k = 1, \dots, K$) with the smallest distances from $\mathbf{z}_0^i$ are selected and their corresponding data at time t , $x_{t,k}^i$ is used for estimation of $x_{t,0}^i$, as shown in Eq. (8). In this study, we set the number of nearest neighbors as 14 ($K = 14$).

$$x_{t,0}^i = \frac{\sum_{k=1}^K x_{t,k}^i}{\sum_{k=1}^K \frac{1}{d_k^i}} \quad (8)$$

Once the missing values are replaced with the output of KNN, the DAE model becomes applicable. The output of the DAE model, $\hat{x} = \mathbf{g}_\theta(\mathbf{f}_\theta(\mathbf{x}))$, is expected to be more accurate than the output of the KNN; however, it is different from the target value because it is trained based on the outputs of KNN, which are roughly estimated initial values. The outputs of the DAE models then replace the outputs of the KNN. This updating process goes on until the difference between the updated value and the previous value is minimized. When there are M missing values, the difference is measured by the following equation, where l is the index of the gas that has been missed.

$$\Delta = \frac{\sum_{l=1}^M \Delta^l}{M} = \frac{1}{M} \sum_{l=1}^M \frac{|\hat{x}_{\text{new}}^l - \hat{x}_{\text{old}}^l|}{|\hat{x}_{\text{new}}^l|} \quad (9)$$

The pseudo-code for the algorithm is shown in Table 2. $\tilde{x}$ is the missed input data and N is the number of data samples.

4. Case study: transformer diagnosis model

The proposed method was verified by examining actual industrial

Table 2

Pseudo-code for the IDAE algorithm.

Algorithm	Impute missing values with IDAE.
Requirements:	Input dataset $\tilde{\mathbf{X}} = \{\tilde{\mathbf{x}}_1, \dots, \tilde{\mathbf{x}}_N\}$ with missing values.
1. Initialize:	
2. $l = \{l_1, \dots, l_N\}$:	Define l where l_n is missing value index for n th data.
3. $\tilde{\mathbf{X}} = \emptyset$:	Dataset of imputed input.
4. for n in N do	
5. Make initial guess $\hat{\mathbf{x}}_{n,old}$	for missing values using KNN.
6. while $\Delta_{new} < \Delta_{old}$ do	
7. for l_n in l do	
8. Predict $\hat{\mathbf{x}}_{n,new}$	corresponding to $\tilde{\mathbf{x}}_n$ using DAE model;
9. end for	
10. Calculate Δ_{new} ;	
11. end while	
12. Append $\hat{\mathbf{x}}_{n,new}$	to $\tilde{\mathbf{X}}$;
13. end for	
14. return imputed matrix $\tilde{\mathbf{X}}$	

data. Three comparative studies were conducted. The first and second case studies were explored for verification of the imputation model for single and multiple missing values, respectively. The last case study compared the diagnostic performance with and without imputation.

4.1. Transformer DGA dataset

The DGA data used for the case studies in this research was provided by the Korea Electric Power Corporation (KEPCO). The data was gathered from more than 8000 transformers over 10 years. This data reflects the intact characteristics of the operating conditions and data gathering conditions. The first rated voltage of the transformers in the data set varies from 22.9 kV to 765 kV; the transformers were manufactured by 49 different companies.

Based on the experts from KEPCO and the literature reports on DGA methods, six primary gases, H_2 , C_2H_2 , C_2H_4 , C_2H_6 , CH_4 , and CO , were selected for analysis. Other gases, such as O_2 , N_2 , and CO_2 , were excluded from the analysis because these gases can be generated from reasons other than degradation. Erroneous data, data with an obviously incorrect value, and data corrupted by oil filtering, were excluded. The total number of data samples in the final study set is 17,850. About 90% of these were used for training, and the remaining 10% was used for testing.

4.2. Comparative studies

4.2.1. Estimation model for a single missing value

For the first comparative study, the accuracy of single missing value estimation will be compared using the proposed method and other previous methods. The comparison algorithms used alongside the proposed method are k-nearest neighbour (KNN) [43], XGBoost (XGB) [44], Light-gbm (LGB) [45], and random forest (RF) [46].

The accuracy of each algorithm is measured using two metrics. First is the normalized root mean square error (NRMSE), which is the root mean square error divided by the mean of the square of the original value. It is defined in Eq. (10).

$$\text{NRMSE} = \sqrt{\frac{\sum_{k=1}^N (\mathbf{x}^k - \hat{\mathbf{x}}^k)^2}{\sum_{k=1}^N (\mathbf{x}^k)^2}} \quad (10)$$

NRMSE alleviates the scale difference of gases, so it allows a fair comparison between gases. However, since NRMSE is vulnerable to outliers, it can give biased results, despite generally good performance. Hence, another metric, correlation coefficient, is adopted to supplement the NRMSE, as shown in Eq. (11). The correlation coefficient measures the linearity between the target values and the estimated values, a higher value indicates better estimation.

$$r = \frac{\sum_{k=1}^N (\mathbf{x}^k - \bar{\mathbf{x}})(\hat{\mathbf{x}}^k - \bar{\hat{\mathbf{x}}})}{\sqrt{\sum_{k=1}^N (\mathbf{x}^k - \bar{\mathbf{x}})^2} \sqrt{\sum_{k=1}^N (\hat{\mathbf{x}}^k - \bar{\hat{\mathbf{x}}})^2}} \quad (11)$$

where $\bar{\mathbf{x}}$ indicates the mean value.

Using these metrics, the estimation results are shown in Table 3. Both NRMSE and correlation coefficient results show that the proposed DAE method is superior to the other methods in estimating the gases, including C_2H_6 , CH_4 , and CO . In the case of C_2H_2 , DAE shows the lowest NRMSE, and the correlation coefficient is the second highest. H_2 and C_2H_4 gases are concentrated at 0 ppm; thus, it seems that overfitting has occurred due to data imbalance. Although its performance is somewhat poor for H_2 and C_2H_4 , it has the best overall performance.

KNN has the lowest performance for more than three gases; it also takes ranks lower for the other gases. This is because KNN is dependant on the nearest data. Therefore, when there is an abrupt increase or decrease around the missing value, which is common in DGA data due to the long intervals between samplings, KNN shows low accuracy. This phenomenon is dominant, especially for gases that are rarely generated, such as H_2 , and C_2H_2 . These gases require high enthalpy to be generated; thus, in most cases they are absent. However, when the requirements are met, they are generated, which is regarded as an abrupt appearance in the KNN model.

XGBoost, light-gbm, and random forest are based on the decision tree. Tree-based methods learn criterion that separate the data into groups with similar properties. The criterion is represented as the branch of the tree, and the more the branch is subdivided, the more elaborate the model obtained. However, with DGA data, the decision tree methods have difficulty subdividing the criterion, because most of the DGA data is close to 0 ppm. However, these tree-based algorithms show generally better results than KNN, because the finely divided models are merged by ensemble learning and avoid the overfitting. The tree-based ensemble models performed generally better than DAE for H_2 and C_2H_4 because of these overfitting prevention characteristics. In particular, RF is not overfitted more than the other algorithms, because it uses a bagging method. However, when the distribution is relatively evenly spread, the performance of DAE is better; this is confirmed for the rest of the four gases.

4.2.2. Imputation for multiple missing values

Here, the performance results of IDAE are compared as the number of missing values increases from one to four. 80 samples are generated for each possible combination. For example, there are 12 possible combinations for two missing values, so 960 samples are generated. In this way, a total of 2480 samples are prepared.

All missing samples were imputed by applying the proposed IDAE. In Table 4, the NRMSE and correlation coefficient between the imputed value and the actual value are summarized. As the number of missing values increases, the performance gradually decreases. This is a natural result, because errors in the single gas imputation model accumulate as the missing values increase.

Comparing the performance decline levels when there is one missing value and four missing values, the NRMSE increased by 0.096 and the correlation coefficient decreased by 0.04. As the number of missing values increases, the NRMSE increases by 0.032 and the correlation coefficient decreased by 0.013, on average. This means that as missing values are added one by one, the performance declines about 3.2% for NRMSE and 1.3% for the correlation coefficient approach. This seems to be very stable because the performance reduction rate according to the number of missing values is very small. To understand the effectiveness of the initial value set by KNN, imputation is performed with and without the KNN method. When the initial value was not set by KNN, the RMSE increased by 0.231, on average, and the correlation coefficient decreased by 0.135, on average.

Table 3

Imputation performance comparison for a single missing value.

Metric	Methods	H ₂	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	CH ₄	CO
NRMSE	DAE	0.532	0.39	0.302	0.165	0.187	0.213
	XGB	0.506	0.525	0.313	0.270	0.372	0.317
	LGB	0.479	0.425	0.262	0.202	0.272	0.222
	RF	0.459	0.425	0.241	0.202	0.204	0.255
	KNN	0.591	0.579	0.275	0.279	0.330	0.226
Correlation coefficient	DAE	0.807	0.904	0.934	0.981	0.974	0.907
	XGB	0.818	0.855	0.944	0.952	0.913	0.820
	LGB	0.824	0.881	0.957	0.974	0.947	0.900
	RF	0.834	0.907	0.961	0.974	0.969	0.883
	KNN	0.730	0.770	0.945	0.951	0.927	0.899

Table 4

Imputation performance of IDAE according to the number of missed DGA values.

Number of missing values	1	2	3	4
NRMSE	0.289	0.313	0.330	0.385
Correlation coefficient	0.919	0.904	0.894	0.879

4.2.3. Diagnostic model with recovered data

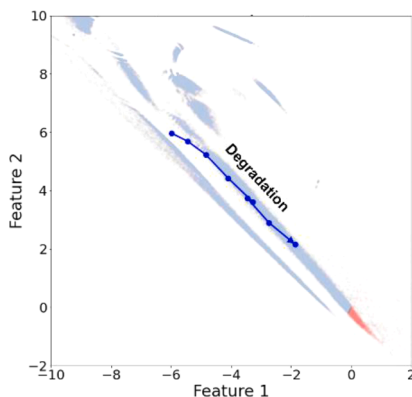
Original and recovered data are used to construct the diagnosis models, and the diagnostic performances are compared. For diagnostic models, two models, IEEE Std. C57.104–2008 and semi-supervised autoencoder with an auxiliary task (SAAT) model, are used. IEEE Std. C57.104–2008 is the standard rule-based diagnostic method and is widely used. The model accepts H₂, C₂H₂, C₂H₄, C₂H₆, CH₄, CO, CO₂, and TDCG gases as input and gives the health status of the transformer with four severity levels.

The SAAT is the AI-based diagnostic model that can classify the normal and fault states of a power transformer by extracting health features that represent the deterioration of the insulation. The SAAT model can visualize the degree of the degradation, so the effectiveness of the recovered data can be inspected visually. The transformer data in the feature space of the SAAT model is shown in Fig. 4. It shows the decision boundary between the normal (sky blue) and the fault area (red). When the transformer degrades, the data in the feature space moves toward the red area. The sample degradation data over time is designated in Fig. 4.

For quantitative analysis of the diagnostic performance, the recall metric is used. The recall metric is defined as the ratio of the true positive to the sum of the true positive and false negatives, as shown in Eq. (12).

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (12)$$

where TP and FN stand for the true positive and the false negative, which indicate correct and incorrect estimation on the faulty state, respectively. A value close to 1 indicates better performance in recall.

**Fig. 4.** Feature space of the SAAT model.

Although there are other metrics, recall is considered a suitable metric for a diagnosis model because in diagnosis, knowing the faulty state correctly is more important, even at the expense of false alarm.

Table 5 shows the recall scores of the IEEE and SAAT models. The same dataset as in the previous section is used. In both the IEEE and SAAT models, the performance is improved after data imputation. In particular, the results of the SAAT model are significantly improved after imputation. The SAAT model examines the concentration and composition ratio of all gases together. On the other hand, in the IEEE model, a limit is set for each gas to evaluate the health grade. This limit is judged as the most serious result amongst them. Therefore, as the number of missing values increases, the SAAT model is more likely than the IEEE model to determine the condition to be normal.

Fig. 5 shows the true data, missing data, and imputed data in the feature space of SAAT. The imputed data is located near the true data around the faulty area; however, the data with missing values departs from it. Also, as the number of missing values increases, the imputed data maintains the data in the faulty area with a slight drift from the true data, but the missed data is greatly affected. As a result, the imputed data is robust to false alarms.

5. Conclusion

In this study, we proposed a new methodology that imputes multiple missing values of DGA through DAE-based iterative imputation. The proposed methodology proceeds via three main steps: (1) preprocessing DGA data, (2) learning the DAE imputation model for a single missing value, and (3) iterative imputation of multiple missing values using the DAE model. The model is validated through three case studies that examined a vast amount of real industrial data.

The first study compares the imputation performance for six types of single gases, using conventional algorithms. The DAE model had the best imputation performance for overall gases; thus, it was adopted as a core imputer in the iterative imputation process. The second study tested the performance of the proposed methodology for imputing multiple missing values. The number of missing values was increased from one to four, and we tested the imputation performance according to the number of missing values. The performance gradually decreased as the number of missing values increased. This is an expected result because the number of unknown variables to be estimated increases. The third study confirmed the difference in diagnostic performance before and after imputation based on existing diagnosis methods for power transformers. The diagnostic methods used in the study are rule-based IEEE Std. C57.104–2008 and an artificial intelligence-based SAAT model.

Table 5

Recalls of the IEEE and SAAT models before and after imputation.

	Imputation	N = 1	N = 2	N = 3	N = 4
IEEE	No	0.852	0.770	0.714	0.671
	Yes	0.985	0.927	0.937	0.893
SAAT	No	0.359	0.130	0.018	0
	Yes	0.829	0.794	0.828	0.692

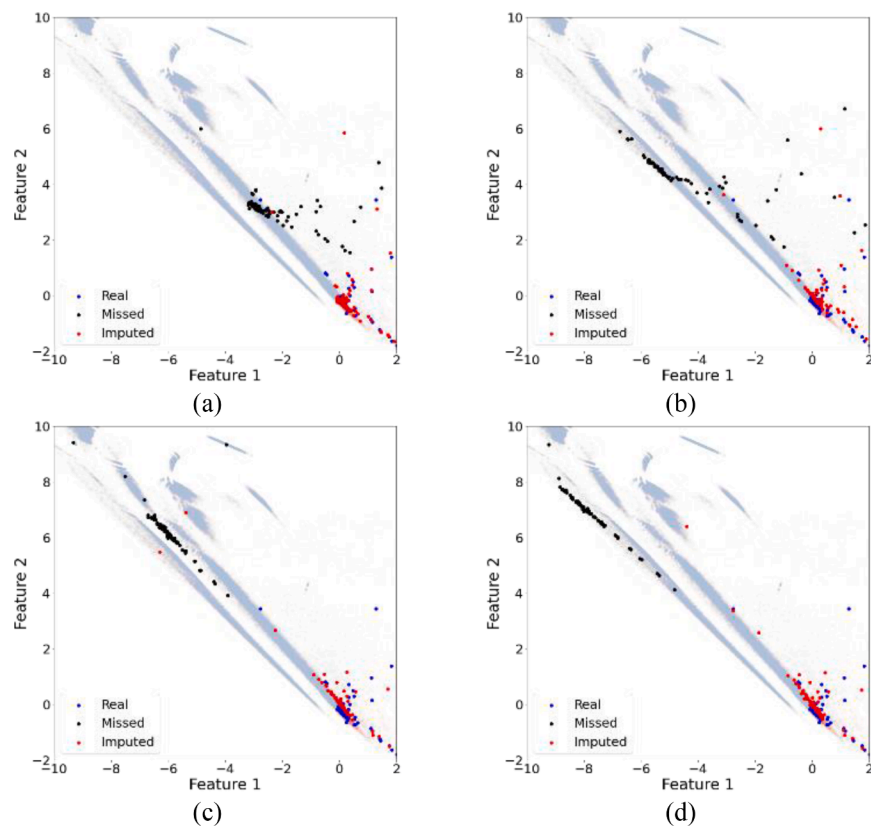


Fig. 5. The results of SAAT diagnosis before and after the imputation: (a) one missing value, (b) two missing values, (c) three missing values, (d) four missing values.

Both methods significantly improved diagnostic accuracy after correction.

Most oil-immersed transformers are managed through DGA. Much research has been conducted to develop accurate diagnosis models; however, prior work has not thoroughly examined the importance of the integrity of the data used in the diagnosis. In particular, DGA data has a long measurement cycle and a small number of data; thus, each data point is meaningful. Therefore, the methodology proposed in this study is expected to further improve the performance of the diagnosis model and greatly reduce post-mortem maintenance that is required when a wrong diagnosis is provided due to missing data.

There are other data issues that should also be considered, such as noise in DGA data. In this study, as field data is used, the noise level cannot be quantified and the estimation results based on the noise level are not thoroughly investigated. Since the noise simulation model requires further research, it will be addressed in future studies.

Due to the limitations of the offline DGA method, attempts have recently been made in industry to gradually convert to an online DGA method. Online methods have the advantage of higher sampling frequency and lower probability of missing data. However, due to the physical limitations of the measurement method, the data accuracy is lower than that of the offline gas chromatography method. In future studies, research for increasing the accuracy of online DGA data will be conducted.

CRedit authorship contribution statement

Boseong Seo: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization. **Jaekyung Shin:** Software, Investigation, Data curation. **Taejin Kim:** Conceptualization, Writing – original draft, Writing – review & editing, Supervision. **Byeng D. Youn:** Supervision, Project administration, Resources, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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